Lecture 18: Matroids and the Greedy Algorithm

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November 5, 2026
601.433/633 Introduction to Algorithms

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- Universe U
- Collection $\mathcal{I} \subseteq 2^U$ (so $I \subseteq U$ for all $I \in \mathcal{I}$). Called *independent sets*
- Weights $w : U \to \mathbb{R}_{\geq 0}$

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Problem: find *max weight* independent set

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So under weights w', max-weight IS = max-weight forest = max-weight spanning tree = min-weight spanning tree (weights w)

- So finding max-weight forest = finding min spanning tree.

Useful Properties of Forests

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Suppose false: no edge in $F_2 \setminus F_1$ can be added to F_1. Let $c_1 = \#$ components in F_1, $c_2 = \#$ components in F_2

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Contradiction. □

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Definition

$(U, \mathcal{I})$ is a *matroid* if the following three properties hold:

1. $\varnothing \in \mathcal{I}$,
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Warmup: In any matroid, the maximal independent sets (called bases) have the same size (called the rank of the matroid).

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  - U a finite set of vectors in $\mathbb{R}^d$
  - $\mathcal{I} = \{F \subseteq U : F \text{ linearly independent}\}$

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  - $\boldsymbol{U}$ a finite set of vectors in $\mathbb{R}^{\boldsymbol{d}}$
  - $\mathcal{I} = \{\boldsymbol{F} \subseteq \boldsymbol{U} : \boldsymbol{F}$ linearly independent$\}$
  - $\varnothing$ linearly independent
  - If $\boldsymbol{F}$ linearly independent and $\boldsymbol{F}' \subseteq \boldsymbol{F}$, then $\boldsymbol{F}'$ linearly independent
  - Augmentation: if $\boldsymbol{F_1}$ linearly independent, $\boldsymbol{F_2}$ linearly independent, and $|\boldsymbol{F_2}| > |\boldsymbol{F_1}| \implies \dim(\text{span}(\boldsymbol{F_1})) = |\boldsymbol{F_1}| < |\boldsymbol{F_2}| = \dim(\text{span}(\boldsymbol{F_2}))$

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Matroids: generalize both graph theory and linear algebra!

- Originally invented by Whitney as an attempt to generalize the concept of "linear independence"

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We'll assume we have independence oracle.

Greedy Algorithm

Kruskal, generalized to matroids (and max weight)!

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$F = \varnothing$
Sort U by weight (largest to smallest)
For each $u \in U$ in sorted order $\{$
  If $F \cup \{u\} \in \mathcal{I}$, add u to F
$\}$
Return F

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- $F = \{f_1, f_2, \ldots, f_r\}$, where $w(f_i) \geq w(f_{i+1})$ for all i (order added by greedy)
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Contradiction! Greedy would add e_z next, not f_j.

Converse

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So for hereditary set systems, matroids exactly characterize when the greedy algorithm works!

Proof

Contradiction. Suppose false $\implies$ $(U, \mathcal{I})$ hereditary but not matroid.

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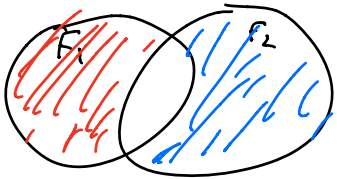
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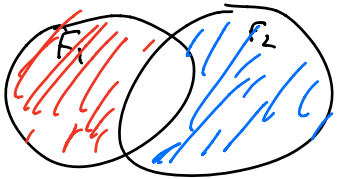
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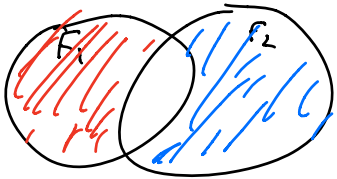
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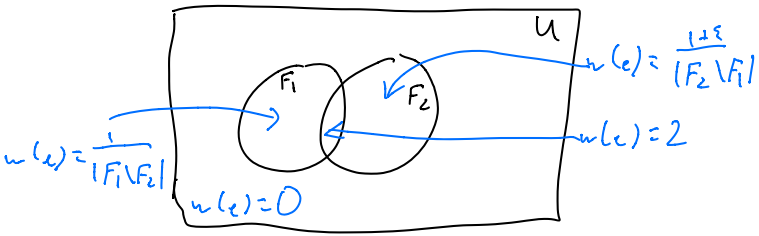
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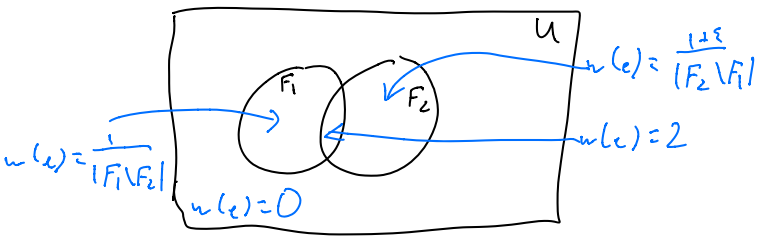
Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.

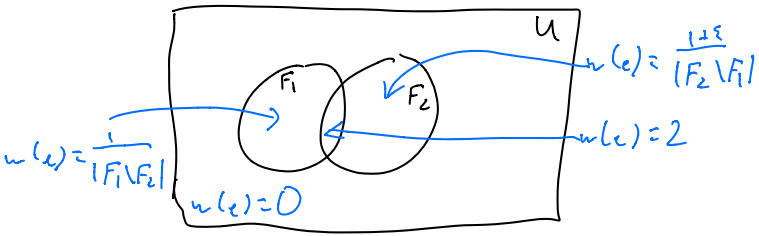


Greedy:

- Adds all of $F_1 \cap F_2$
- Adds all of $F_1 \setminus F_2$
- Can't add any of $F_2 \setminus F_1$

Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



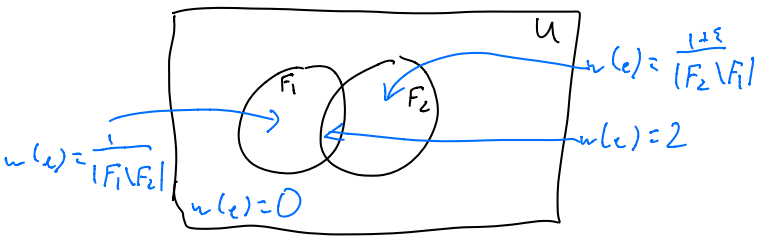
Greedy:

- Adds all of $F_1 \cap F_2$
- Adds all of $F_1 \setminus F_2$
- Can't add any of $F_2 \setminus F_1$

$$w(\text{greedy}) = 2|F_1 \cap F_2| + |F_1 \setminus F_2| \frac{1}{|F_1 \setminus F_2|}$$
$$= 2|F_1 \cap F_2| + 1$$

Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



Greedy:

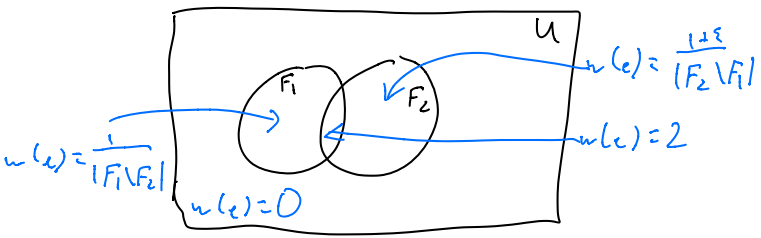
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$$w(\text{greedy}) = 2|F_1 \cap F_2| + |F_1 \setminus F_2| \frac{1}{|F_1 \setminus F_2|}$$
$$= 2|F_1 \cap F_2| + 1$$

$$w(F_2) = 2|F_1 \cap F_2| + |F_2 \setminus F_1| \frac{1+\epsilon}{|F_2 \setminus F_1|}$$
$$= 2|F_1 \cap F_2| + 1 + \epsilon$$

Proof (cont'd)

Use fact that $\frac{1}{|F_1 \setminus F_2|} > \frac{1+\epsilon}{|F_2 \setminus F_1|}$ to define weights.



Greedy:

- Adds all of $F_1 \cap F_2$
- Adds all of $F_1 \setminus F_2$
- Can't add any of $F_2 \setminus F_1$

$$w(\text{greedy}) = 2|F_1 \cap F_2| + |F_1 \setminus F_2| \frac{1}{|F_1 \setminus F_2|}$$
$$= 2|F_1 \cap F_2| + 1$$

$$w(F_2) = 2|F_1 \cap F_2| + |F_2 \setminus F_1| \frac{1+\epsilon}{|F_2 \setminus F_1|}$$
$$= 2|F_1 \cap F_2| + 1 + \epsilon$$

Greedy not optimal: contradiction!